

Abhishek Mandal

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Summary

Full-stack software engineer with **2 production internships** across healthcare and e-commerce, building and shipping systems that process **4,000+ monthly transactions** at scale. Proven ability to identify bottlenecks, design scalable backend architectures, and deliver measurable outcomes, cut reconciliation time by **40%**, improved financial traceability by **60%**, and reduced auth abuse by **90%+**. Proficient in **TypeScript, Node.js, PostgreSQL, React**, and **Next.js**, with hands-on experience in **RBAC design, RESTful API architecture, database query optimisation**, and third-party integrations (Razorpay, Clerk, Cloudinary). Comfortable reasoning about **system design trade-offs**, from schema normalisation and indexing strategies to middleware layering and rate limiting.

Education

Indian Institute of Information Technology (IIIT), Ranchi

2022 – 2026 | CGPA: 7.15

Bachelor of Technology (B.Tech) in Electronics and Communication Engineering

Experience

Betterway Ayurveda — SDE Intern

Jan 2026 – Present | Gurugram, Haryana

TypeScript · Node.js · Express.js · React · Next.js · Expo · PostgreSQL · Redux · Retool

- Identified fragmented billing across consultations, medicines, and Panchkarma as a reconciliation bottleneck; designed and built a unified Packages Billing System, cutting monthly reconciliation time by 40% across 500+ patient records in the EMR.
- Designed a session-level Payment Ledger that captures transaction metadata, timestamps, and service context per patient visit, improving financial traceability by 60% and producing audit-ready records across 4,000+ monthly transactions.
- Architected a scalable RBAC middleware layer across all critical EMR API endpoints, defining role-permission mappings for doctors, admins, and support staff, eliminating unauthorized access and establishing a reusable auth pattern across the codebase.
- Built a real-time Notification & Escalation System with structured state tracking (open → in-progress → resolved), cutting escalation resolution time by 35% and giving teams a clear audit trail and priority queue.
- Reduced manual reporting effort by 50% by delivering custom Retool dashboards with complex SQL aggregations and JS transformations, enabling ops and finance teams to self-serve on EMR data without engineering support.

CVNT Fashion & Apparel – SDE Intern

Oct – Dec 2025 | Remote

Node.js · Express.js · Next.js · PostgreSQL · Supabase · Tailwind CSS

- Built a hybrid authentication system combining Google OAuth and OTP-based verification, eliminating race condition failures that caused silent login drops under concurrent sessions.
- Implemented OTP lifecycle management with atomic PostgreSQL RPCs and per-IP rate limiting, cutting OTP abuse attempts by 90%+ under peak traffic.
- Delivered platform-wide UX improvements and integrated Google Tag Manager with structured event schemas, enabling the marketing team to track conversion funnels without further engineering involvement.

Projects

NK Consultancy – CA Consultancy Platform 🔗

Jul – Sep 2025

React · Node.js · Express.js · MongoDB · Tailwind CSS · Clerk · Razorpay · Cloudinary

- Built a MERN consultancy platform with a JSON-driven dynamic form engine rendering service-specific forms from metadata, supporting 8 live services across 12 clients with no frontend code changes per new service.
- Secured the backend with Clerk JWT auth and custom RBAC middleware for Admin/Client isolation; integrated Razorpay with HMAC-SHA256 signature verification ensuring 100% transactional integrity.
- Built a cloud document pipeline with Multer + Cloudinary for multi-format uploads and an admin dashboard with Recharts visualizing revenue trends and client conversion data.

Tap In – Event Ticketing & Management Platform 🔗

Jun – Jul 2025

React.js · Node.js · Express.js · MongoDB · Razorpay · Gemini API · Clerk · Cloudinary

- Architected a MERN ticketing platform with Clerk webhook-driven identity sync to MongoDB and RBAC middleware enforcing Organizer/Attendee role separation on all RESTful endpoints.
- Engineered an end-to-end QR ticketing lifecycle, high-entropy hashes generated post Razorpay HMAC-SHA256 payment verification, with html5-qrcode enabling real-time server-side attendee validation at venue.
- Integrated **Gemini API** for AI-assisted event drafting and a Cloudinary/Multer CDN pipeline to offload image processing from the main server.

Technical Skills

Languages: TypeScript, JavaScript (ES6+), C++, C, SQL

Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS, Shadcn/ui, Expo, Material UI, HTML5, CSS3

Backend: Node.js, Express.js, RESTful APIs, JWT, OAuth 2.0, Prisma ORM, Zod, WebSockets, Microservices

Databases: PostgreSQL (indexing, query optimisation, transactions, ACID), MongoDB, Supabase

Caching & Queues: Redis (caching, pub/sub, session management), Kafka

System Design: LLD patterns (Singleton, Observer, Strategy, Factory), SOLID principles, Rate Limiting, Idempotency

DevOps & Cloud: Docker, AWS S3, Vercel, Render, Cloudinary, Git, GitHub Actions

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, OS Concepts, HTTP/REST

Tools: Postman, Retool, Razorpay, OpenAI API, Gemini API, Google Tag Manager

Achievements

- Achieved **LeetCode Knight** badge with a contest rating of **1850+**, ranking in the **top 4%** globally, 500+ problems solved ([Profile](#)).
- Grew a CA practice through marketing and operations improvements; built a full-stack consultancy platform that added **52 clients** and contributed to **30% annual revenue growth**.
- Secured **Rank 3** in the college's internal Smart India Hackathon round as part of a 6-member team.